THE ASSAM
ROYAL GLOBAL UNIVERSITY
GUWAHATI

ROYAL SCHOOL OF ENVIRONMENTAL AND EARTH SCIENCES

(RSEES)

DEPARTMENT OF GEOGRAPHY AND GEOINFORMATICS

M.A/M.Sc. in Geography

Postgraduate Programme as per NEP, 2020

W.E.F

AY - 2026– 27

STRUCTURE OF THE SYLLABUS FOR 2 YEAR PG PROGRAMME

SCHOOL NAME - Royal School of Environment and Earth Sciences (RSEES)
DEPARTMENT NAME - Department of Geography
PROGRAMME NAME - M.A./ M.Sc. Program

1st SEMESTER			
COURSE CODE	**COURSE TITLE**	**CREDIT**	**L-T-P**
GEO164C101	Advanced Geomorphology	4	4-0-0
GEO164C102	Climatology and Biogeography	4	4-0-0
GEO164C103	Statistical Techniques in Geography	4	4-0-0
GEO164C104	Cartographic Techniques	4	4-0-0
GEO164C115	Cartographic Techniques in Geo-environmental Mapping	4	0-0-8
	Swayam Course (Upcoming)	3/4	
	TOTAL CREDIT FOR 1st SEMESTER		**20+3/4**

2nd SEMESTER			
COURSE CODE	**COURSE TITLE**	**CREDIT**	**L-T-P**
GEO164C201	Economic Geography	4	4-0-0
GEO164C202	Evolution of Geographical thought	4	4-0-0
GEO164C203	Geography of Rural Development	4	4-0-0
GEO164C204	Urban Geography	4	4-0-0
GEO164C215	Traditional and Digital Cartographic Techniques	4	0-0-8
	Swayam Course (Upcoming)	3/4	
	TOTAL CREDIT FOR 2nd SEMESTER		**20+3/4**

TOTAL CREDIT FOR YEAR 1= 46/48

3rd SEMESTER			
COURSE CODE	**COURSE TITLE**	**CREDIT**	**L-T-P**
GEO164C301	RS, GIS, GPS: Principles and applications	4	4-0-0
GEO164C302	Social and Cultural geography	4	4-0-0
GEO164C303	Research Methodology in Geography	4	4-0-0
GEO164C304	Disaster Management	4	4-0-0
GEO164C325	Report writing on Environmental issues	4	
	TOTAL CREDIT FOR 3rd SEMESTER	**20**	

or 3rd Semester (For students with 3rd & 4th Semester Research)

GEO164C321	**Research Project Phase – 1**		20

Course Work with Research, (Course Work=12 Credit, Research=8 Credits)

COURSE CODE	**COURSE TITLE**	**CREDIT**	**L-T-P**
GEO164C301	RS, GIS,and GPS: Principles and applications (Remote Sensing)	4	4-0-0
GEO164C302	Social and Cultural Geography	4	4-0-0
GEO164C303	Research Methodology in Geography	4	4-0-0
GEO164C324	Survey/Research I	8	
	Total		**20**

4th SEMESTER

COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P
GEO164C401	Population and Settlement geography	500	4	4-0-0
GEO164C402	Political geography	500	4	4-0-0
GEO164C403	Biodiversity Conservation	500	4	4-0-0
GEO164C404	Geography of Tourism	500	4	4-0-0
GEO164C425	Report Writing on Socio-Economic Issues	500	4	
Or 4th Semester (For students with 3rd & 4th Semester Research)				
GEO162C421	**Research Project Phase – 2** Dissertation only (for students who have completed only coursework of 20 credits in 3rd semester/for 'Coursework only' in lieu of research)		20	
Course Work with Research, (Course Work=8 Credit, Research=12 Credits)				
GEO164C401	Population and Settlement geography		4	4-0-0
GEO164C404	Political Geography		4	4-0-0
GEO164C426	Survey/Research II		12	
Total Credit for 4th Semester			20	
TOTAL CREDIT FOR YEAR 2 = 40				

M.A./M. Sc. Course in Geography: Semester-I

Paper I Core Course	ADVANCED GEOMORPHOLOGY			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO164C 101

Course Objectives: *The pivotal point of this course is to make students familiar with the fundamental concepts of geomorphology, including geomorphic structures, Earth's interior and composition, and the evolution of landforms. The course emphasizes sustainability and resilience, aligning with the United Nations Sustainable Development Goals (SDGs), particularly* ***SDG 13: Climate Action****,* ***SDG 11: Sustainable Cities and Communities****,* ***SDG 6: Clean Water and Sanitation****, and* ***SDG 15: Life on Land****.*

Course Outcomes:

By the end of this course the students will be able to:

CO No.	Course Outcome	Blooms Taxonomy Level
CO1.	Recall the principles and basic concepts of geomorphology and its recent trends, contributing to **SDG 4: Quality Education**.	BT 1
CO2.	Outline the roles of structure, process, and time in shaping landforms and interpret geomorphological maps, supporting **SDG 11: Sustainable Cities and Communities**.	BT 2
CO3.	Apply geomorphological knowledge in geographical research, advancing **SDG 13: Climate Action**.	BT 3
CO4.	Distinguish between mechanisms that control geomorphic processes and analyze natural and anthropogenic impacts on landforms, aligned with **SDG 11: Sustainable Cities and Communities** and **SDG 15: Life on Land**.	BT 4
CO5.	Evaluate the functioning of Earth systems in real time with sustainability perspectives, reinforcing **SDG 13: Climate Action** and **SDG 15: Life on Land**.	BT 5

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Advanced Concepts in Process Geomorphology: System dynamics and feedback mechanisms, Rate laws and equilibrium concepts; Weathering Processes and Rates: Chemical and Physical weathering processes; Multicyclic & Polygenetic Evolution of Landscape, Concept of threshold.	12

Unit 2	Plate Tectonic: Concepts of related views; Earth movements: Epeirogenesis, orogenesis, isostasy; Forces of crustal instability: Earthquake and Vulcanicity; Evolution of Landforms: Fluvial Landforms, Drainage Pattern Evolution; Karst Landforms: Erosional and Depositional Features.	12
Unit 3	Glacial and Periglacial dynamics and erosion processes, Glacial landform development; Coastal Landform evolution: Processes and Features; Aeolian processes and landforms, Drainage systems in arid regions; Climate change impacts.	12
Unit 4	Remote Sensing in Geomorphology; Geomorphological Hazards: Landslide, Flood analysis and prediction, Coastal erosion, Risk assessment; Human impacts on landscapes; Restoration approaches and techniques, Case Studies.	12
	Total	**48**

Text Books:

1. Hart, M. G. (1986): Geomorphology Pure and Applied, George Allen & Unwin, London
2. Dayal, P. (2nd edition), (1996): A Textbook of Geomorphology, Shukla Book Depot, Patna
3. Savindra Singh (Rep. 2011): Geomorphology, Prayag Pustak Bhawan, Allahabad.
4. Strahler A. H and Strahler, A. N. (1992): Modern Physical Geography, John Wiley, New York.
5. Thornbury, W. D. (Rep.2011): Principles of Geomorphology, John Wiley and Sons, New York.

Reference Books:

1. Ahmad, E., 1985: Geomorphology, Kalyani Publishers, New Delhi
2. Derbishire, E. (ed), 1976: Geomorphology and Climate, Wiley, London.
3. Fairbridge, R.W. (ed), 1968: Encyclopedia of Geomorphology, Reinhold, New York
4. Gregory, K, J. and Walling, D.E., 1973: Drainage Basin- Form and Process, Edward Arnold, London
5. Goudie, Andrew, et. Al. (eds), 1981: Geomorphological Techniques, George Allen & Unwin, London.
6. Kale, V. S. and Gupta, A. (Rep.2011): Introduction to Geomorphology, Orient Longman, Calcutta.
7. Ollier, C. D. (1981): Tectonics and Landforms, Longman, London.
8. Spark B .W. (1972): Geomorphology, Longman, New York.
9. Young, A. (1972): Slope, Longman, New York, Oliver and Boyd, Edinburgh

Paper II Core Course	CLIMATOLOGY AND BIOGEOGRAPHY			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO164C102

Course Objectives:

The course aims to illustrate the atmospheric elements, processes, and resultant weather and climates, as well as the impact of climate systems on planet Earth and its associated processes. It emphasizes sustainability and resilience, aligning with SDG 13: Climate Action for climate adaptation and mitigation, SDG 11: Sustainable Cities and Communities for urban climate resilience, and SDG 7: Affordable and Clean Energy for renewable energy planning.

Course Outcomes:

By the end of this course the students will be able to:		
CO No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	Define the elements of weather and climate and their impacts at different scales, contributing to **SDG 13: Climate Action**.	**BT 1**
CO2.	Demonstrate and interpret weather maps, supporting **SDG 11: Sustainable Cities and Communities** through disaster preparedness and urban climate planning.	**BT 2**
CO3.	Develop an understanding of climatic aspects and their bearing on planet Earth, reinforcing **SDG 13: Climate Action**.	**BT 3**
CO4.	Distinguish world climatic regimes, aligned with **SDG 13: Climate Action** and global sustainability perspectives.	**BT 4**
CO5.	Assess the monsoon and its variability, global climate change, and its consequences, directly addressing **SDG 13: Climate Action** and **SDG 11: Sustainable Cities and Communities**.	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	The field of Climatology and its subdivisions, Factors affecting weather and climate, Composition and vertical structure of the atmosphere- physical properties, chemical composition and temperature changes; Insolation, factors affecting insolation, latitudinal and seasonal variation of insolation and the heat budget. Distribution of temperature, Inversion of temperature, Pressure measurement and units, Factors affecting air pressure, wind, Geostrophic wind, Gradient wind, Pressure systems and air circulation, Models of general circulation, Local winds and Jet stream	10
Unit 2	Air mass and fronts: types, characteristics and their influence on weather and climate, Factors affecting evaporation, atmospheric lapse rate and stability; Changes of state of water, factors affecting Condensation; Classification of world climate of Koppen and Thornthwaite. Monsoon climate: origin, mechanism of development and distribution. Climatic disturbances: cyclone, anticyclone,	14

	drought, cloud burst, El-Nino, La-Nina; Techniques of weather forecasting, Causes of global warming and climate change	
Unit 3	Nature, scope and subject matter of biogeography, ecology, ecosystem, trophic level, food chain and energy flow in ecosystem, nutrient cycle. Classification, characteristics, and global distribution of biomes; phytogeographical and zoogeographical regions of the world with special reference to India; factors influencing the distribution and adaptation of plants and animals.	10
Unit 4	Biodiversity, biodiversity hot spot, conservation of biodiversity and mechanism: national park, marine national park, wildlife sanctuary, reserve forest, bird sanctuary; with special reference to India. Conservation of the environment, important environmental days, EIA, UNFCC, UNCED, environmental policies in India.	14
	Total	**48**

Text Books:

1. Barry, R. G. and Chorley, R. J. (1971): Atmosphere, Weather and Climate, Methuen Co. Ltd, London.
2. Critchfield, H. J. (Rep.2010): General Climatology, Prentice Hall, New Delhi.
3. Lal, D. S. (2003): Climatology, Sharda Pustak Bhawan, 11 , University Road, Allahabad-211002
4. Rob Vandenbarg (2009): Evaluating Climate Change and Development.
5. Singh, S. (Rep.2011): Climatology, Prayag Pustak Bhawan, Allahabad

Reference Books:

1. Lutgens F. K., Tarbuck E. J. and Tasa D., (2009): The Atmosphere: An Introduction to Meteorology, Prentice-Hall, Englewood Cliffs, New Jersey
2. Oliver J. E. and Hidore J. J., (2002): Climatology: An Atmospheric Science, Pearson Education, New Delhi.
3. Stringer, E. N., 1982: Hoirn, L. A., (1980): An Introduction to Climate, International Series.
4. Trewartha, G. T. and Horn, L. A., (1980): An Introduction to Climate, International Series.
5. Weiesner, C. J.: Hydrometeorology, Chapman & Hall Ltd.

Paper III Core Course	STATISTICAL TECHNIQUES IN GEOGRAPHY L-T-P-C: 4-0-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEO164C103

Course Objectives: *This paper provides an understanding of the pure and applied nature of Geography along with the key elements in the discipline. It emphasizes the role of statistical methods and quantitative techniques in evidence-based decision making, supporting* ***SDG 4: Quality Education*** *through skill development,* ***SDG 9: Industry, Innovation and Infrastructure*** *through data-driven innovation, and* ***SDG 11: Sustainable Cities and Communities*** *through urban and regional planning.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	*Define the statistical methods and quantitative techniques used in Geography, contributing to* ***SDG 4: Quality Education****.*	**BT 1**
CO2.	*Demonstrate and interpret various methods of data collection, tabulation, interpretation, and analysis, supporting* ***SDG 9: Industry, Innovation and Infrastructure****.*	**BT 2**
CO3.	*Identify the importance of data in geography for sustainable development, aligned with* ***SDG 11: Sustainable Cities and Communities****.*	**BT 3**
CO4.	*Analyse data through tabulation, sample size determination, and other methods by handling data in the field, reinforcing* ***SDG 9: Innovation*** *and* ***SDG 11: Urban Planning****.*	**BT 4**
CO5.	*Interpret data and validate hypotheses to support evidence-based geographical research, advancing* ***SDG 4: Quality Education*** *and* ***SDG 9: Innovation****.*	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Quantitative and qualitative techniques; Significance and limitations of quantitative techniques; Descriptive and inferential statistics; Levels of measurement; Data sources and acquisition techniques; Sample and sampling techniques; Geographic data matrix.	12
Unit 2	Measures of central tendencies (Mean, Median and Mode); Measures of dispersion (Range, Quartile Deviation, Mean Deviation, Standard Deviation; Coefficient of variation); Concept of spatial mean and median centres and standard distance and its uses, Nearest Neighbour Analysis (NNA), Inferential statistics – Chi-square (c^2) Analysis, Concept of ANOVA and F-test;	12
Unit 3	Correlation and regression analysis (simple and multiple), Regression residual mapping, Parametric tests - t -test; Nonlinear relationships - Exponential and power function types; Theoretical distributions: Normal, Poisson and Binomial.	12
Unit 4	Basics of matrix algebra: Matrices - definition, types; minors and co-factors, determinant of a square matrix, inverse, adjoint, solutions of linear equations; Linear algebra in developing multivariate regression models;	12
	Total	**48**

Note: Computer / calculator based compulsory home assignments may be given for various units. Scientific calculator may be permitted in the examination hall for this paper.

Text Books:

1. Gregory, S., 1978: Statistical Methods and the Geographer, Longman, London.
2. Hammond R and P. S. McCullagh, 1974: Quantitative Techniques in Geography: An Introduction, Clarendan Press, Oxford.
3. Johnston R. J., 1973: Multivariate Statistical Analysis in Geography, Longman, London.
4. King, L. J., 1969: Statistical Methods in Geographical Studies, London.
5. Mahmood, A., 1977: Statistical Methods in Geographical Studies, Concept Publications, Delhi.
6. Maurice Yeats, 1974: An Introduction to Quantitative Analysis in Human Geography, McGraw, Hill, New York.
7. Paul, S. K., 1998: Statistics for Geoscientists, Tata McGraw Hill, New Delhi.
8. Robinson, G. M., 1998: Methods and Techniques in Human Geography, John Wiley & Sons, Chichester.
9. Sarkar, A. 2013: Quantitative Geography: Techniques and Presentations. Orient BlackSwan Private Ltd., New Delhi

Reference Books:

1. David Unwin, 1981: Introductory Spatial Analysis, Methuen, London.
2. John P.Cole and Cuchlaine A. M. King, 1968: Quantitative Geography, John Wiley, London.
3. Koutsoyiannis, 1973: Theory of Econometrics, Mcmillan, London.
4. Peter Haggett, Andrew D. Cliff, & Allan Frey, 1977: Location Methods Vol. I and II, Edward Arnold, London.
5. Taylor P.J., 1983: Quantitative Methods in Geography: An Introduction to SpatialAnalysis, Waveland Press, Boston Publishers.

Paper IV Core Course	CARTOGRAPHIC TECHNIQUES			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO164C104

Course Objectives: *This course focuses on helping students understand the science and art of map-making, including the basics of maps, map scales, and their varied types, along with diagrammatic representation of geographical data. It emphasizes the role of cartography in sustainable development, aligning with* ***SDG 9: Industry, Innovation and Infrastructure*** *for geospatial innovation,* ***SDG 11: Sustainable Cities and Communities*** *for urban and regional planning, and* ***SDG 15: Life on Land*** *for ecosystem monitoring and conservation.*

Course Outcomes:

By the end of this course the students will be able to:		
CO No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	Define maps and their types, map scales, and profile mapping, contributing to **SDG 4: Quality Education**.	**BT 1**
CO2.	Interpret different projection systems and digital mapping analysis, supporting **SDG 9: Innovation and Infrastructure**.	**BT 2**
CO3.	Construct graphs, charts, cartograms, and thematic maps based on socio-economic, cultural, and climatic data, aligned with **SDG 11: Sustainable Cities and Communities**.	**BT 3**
CO4.	Conclude the importance of maps for regional development and decision-making, reinforcing **SDG 11: Sustainable Cities and Communities**.	**BT 5**
CO5.	Evaluate different aspects of surveying and levelling using varied techniques and equipment, advancing **SDG 9: Innovation** and **SDG 15: Life on Land**.	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Cartography: The science and art of map making; Nature and scope of Cartography; Earth as a cartographic problem, Scales, Coordinate systems, distance and directions in maps; Developable and developed surfaces - types and properties; Map projection systems and types; Importance of cartography in geography; Digital cartography-data sources for digital mapping and analysis.	10
Unit 2	Map Projection: Basic concepts, classification, basic Principles of construction of zenithal, conical and cylindrical groups of map projections and choice of map projections.	14
Unit 3	Principles of mapping and base map preparation; Concept of generalization; Map design and layout; Thematic map and its classes; Choropleth and isopleth maps; Map reading and analysis; Representation of socio-economic and climatic data	12
Unit 4	Basic idea of geodetic and plane surveying; Principles and techniques of surveying by: Plane Table, Prismatic Compass, Transit Theodolite, Levelling by Dumpy Level / Auto level, Total Station (Introductory idea)	12
	Total	**48**

Text Books:

1. Misra, R. P., et al., 2014: Fundamentals of Cartography, Concept Publishing co. New Delhi
2. Monkhouse, F. J., and Wilkinson, H. R., 1989: Maps and Diagrams, B. I., Publications Pvt. Ltd., New Delhi-59

3. Robinson, A. H., et al., 1995: Elements of Cartography, 6th Edition, John Wiley & Sons, New York.
4. Steers, J. A., 1965: An Introduction to the Study of Map Projection, University of London, London.
5. Talukder, S., 2008: Introduction to Map Projections, EBH Publishers (India), Guwahati.

Reference Books:

1. Cuff, D. J. and Mattson, M. T., 1982: Thematic Maps: Their Design and Production, Methuen, New Work
2. Singh, R. L.: Elements of Practical Geography, Kalyani Publishers, New Delhi.
3. Sing, R. L. and Singh Rana, P. B., 1998: Elements of Practical Geography, Kalyani Publishers, New Delhi.
4. Saha, P. K. and Basu, P., 2010: Advanced Geography Practical – A Laboratory Manual, Books and Allied (P) Ltd., Kolkata.

Paper V Core Course	Cartographic Techniques in Geo-environmental Mapping			Subject Code:
	L-T-P-C: 0-0-8-4	Credit Units: 4	Scheme of Evaluation: (P)	GEO164C115

Course Objectives: The course aims at increasing the practical knowledge of the students by engaging them with geospatial tools and applications. It emphasizes hands-on learning in environmental mapping and spatial analysis, aligning with **SDG 13: Climate Action** for climate resilience, **SDG 15: Life on Land** for biodiversity and ecosystem conservation, and **SDG 11: Sustainable Cities and Communities** for sustainable urban and regional planning.

Course Outcomes:

By the end of this course the students will be able to:		
CO No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	Define the principles and concepts involved in Practical Geography, contributing to **SDG 4: Quality Education**.	**BT 1**
CO2.	Classify the nature, characteristics, and sources of geospatial data, supporting **SDG 9: Industry, Innovation and Infrastructure**.	**BT 2**
CO3.	Develop technical skills and capabilities in geospatial applications, reinforcing **SDG 11: Sustainable Cities and Communities**.	**BT 3**
CO4.	Simplify the application of concepts related to Geomorphology, Climatology, and Population Geography, aligned with **SDG 13: Climate Action** and **SDG 15: Life on Land**.	**BT 4**
CO5.	Inspect geospatial tools and technologies to create and analyse geospatial data for natural resource assessments, planning, and management, directly addressing **SDG 15: Life on Land** and **SDG 11: Sustainable Cities and Communities**.	**BT 4**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Analysis of Slope by Wentworth's method - **1 Exercise** , Profile drawing (Serial, Superimposed, Composite, Projected) - **2 Exercises**, Drainage basin delimitation, Ordering of streams, calculation of bifurcation ratio, length ratio, drainage frequency and density, computation of basin circularity ratio- **1 exercise**	4
Unit 2	Verification of law of drainage morphometry **- 3 Exercises**, drawing of Hypsometric curve and computation of Integral **-1 Exercise**	4
Unit 3	Preparation of rainfall variability map (Assam and Rajasthan) **- 2 Exercises**, drawing of hythergraph, climograph and ergograph - **3 Exercises**, rainfall frequency analysis, rainfall dispersion graph, water deficiency and surplus graph- **2 Exercises**, weather chart interpretation- **1 Exercise**	8
Unit 4	Mapping of Population distribution, density and concentration in the World and India, Population growth trend and projection in the World and India, Population-Resource regions in the world- **9 Exercises**	8
	Total	**24**

Reference Books:

1. Weiesner, C. J.: ***Hydrometeorology***, Chapman & Hall Ltd.
2. Gregory, K, J. and Walling, D.E., 1973: Drainage Basin- Form and Process, Edward Arnold, London
3. Goudie, Andrew, et. Al. (eds), 1981: Geomorphological Techniques, George Allen & Unwin, London.
4. Woods, R., 1979**: *Population Analysis in Geography***, Longman, London.

M.A./M. Sc. Course in Geography: Semester-II

Paper I Core Course	ECONOMIC GEOGRAPHY			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO164C201

Course Objectives: *This course focuses on the basic concepts of economic geography and its associated patterns and processes of the prime economic activities of the world. It emphasizes the spatial organization of economies and their role in sustainable development, aligning with* ***SDG 8: Decent Work and Economic Growth*** *for employment and productivity,* ***SDG 9: Industry, Innovation and Infrastructure*** *for industrial and infrastructural development, and* ***SDG 10: Reduced Inequalities*** *for equitable distribution of resources.*

Course Outcomes:

By the end of this course the students will be able to:		
CO No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	Define concepts and ways on how geographical aspects organise economic space, contributing to **SDG 8: Decent Work and Economic Growth**.	**BT 1**
CO2.	Compare different sectors of the economy and arrive at logical conclusions regarding their importance in national development, aligned with **SDG 9: Industry, Innovation and Infrastructure**.	**BT 2**
CO3.	Identify the principles and significance of economic geography, reinforcing **SDG 8: Economic Growth** and **SDG 9: Sustainable Industry**.	**BT 3**
CO4.	Discover new insights on the relevance of economy and geography in addressing contemporary problems, supporting **SDG 10: Reduced Inequalities**.	**BT 4**
CO5.	Appraise regional variations in economic scenarios and resource distribution patterns, directly addressing **SDG 8: Decent Work**, **SDG 9: Infrastructure**, and **SDG 10: Social Inclusion**.	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Nature and scope of Economic Geography, its evolution and theoretical development; Approaches in Economic Geography, Types of economic activities and factors of location influencing the activities	10
Unit 2	Economic Geography of Agriculture, Place of agriculture in global economy, Agricultural System and Regions of the World and India, Types of agriculture, Concept of crop combination and its methods of delineation (Weaver, Doi and Rafiullah), diversification and rotation, Von Thunen Model of Land Use Planning.	14
Unit 3	Classification of Industries, Factors of Industrial location, , Theories of Industrial Location- A. Weber, E.M. Hoover, A. Losch, A. Pred and D. M. Smith, Industrial Regions: World and India, Industrial Development experience in India and Japan, World Distribution of ICT (Information and Communication Technology) .	10
Unit 4	Economic Geography of Resources; its global pattern of distribution and production: water, coal, petroleum, forest and nuclear, International trade of Food grains and cash crops, Special Economic Zones, Theories of development: Myrdal and Rostow, Level of economic development on global perspective.	14
	Total	**48**

Text Books:

- Alexander J. W., 1963: Economic Geography, Prentice-Hall Inc., Englewood Cliffs, New
- Coe, N. M., Kelly, P. F., & Yeung, H. W. (2019). *Economic geography: a contemporary introduction*. John Wiley & Sons.
- Combes P., Mayer T. and Thisse J. F., 2008: Economic Geography: The Integration of Introduction, Wiley-Blackwell.
- Hodder B. W. and Lee Roger, 1974: Economic Geography, Taylor and Francis.

Reference Books:

1. Gunner Alexanderson, 1988: ***Geography of Manufacturing***, Prentice-Hall Inc, (E.E. Edition)
2. Isard, W., 1956: ***Location and Space Economy***, MIT Press, Cambridge.
3. Losch, A., 1954: ***The Economics of Location***, New Haven.
4. Smith, David M., 1981: ***Industrial Location: An Economic Geographical Analysis***, Wiley, New York
5. Symons, L., 1979: ***Agricultural Geography***, West view Press, Colorado
6. Thomas, R. S. and Corbin, P. B., 1974: ***The Geography of Economic Activity***, McGraw Hill, New York

Paper II Core Course	EVOLUTION OF GEOGRAPHICAL THOUGHT			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO164C202

***Learning objective**: The objective of this course is to make students explore the chronology of development of geography through the contributions of varied scholars, approaches, and schools, along with major themes and components of the discipline. It emphasizes the role of geographical thought in shaping sustainable development, aligning with **SDG 4: Quality Education** for knowledge building, **SDG 11: Sustainable Cities and Communities** for applying geographical perspectives to urban and regional planning, and **SDG 13: Climate Action** for addressing global environmental challenges*.

Learning Outcomes:

After successful completion of the course the students will be able to:		
CO No.	**Course Outcome**	**Blooms Taxonomy Level**
CO 1.	Define the various parameters and components of Geography, contributing to **SDG 4: Quality Education**.	**BT 1**
CO 2.	Interpret the chronological development of the subject of geography, reinforcing **SDG 4: Quality Education** and sustainability awareness.	**BT 2**
CO 3.	Identify the contributions made by different schools of geography, aligned with **SDG 4: Quality Education** and **SDG 11: Sustainable Cities and Communities**.	**BT 3**
CO 4.	Discover the physical and humanistic perspectives and their dimensions in Geography in relation to physical and cultural surroundings, supporting **SDG 11: Sustainable Cities and Communities** and **SDG 13: Climate Action**.	**BT 4**
CO 5.	Explain real-world issues with a geographical perspective, directly addressing **SDG 11: Sustainable Cities and Communities** and **SDG 13: Climate Action**.	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Place of geography in the classification of knowledge: Defining the field of geography, relation of geography with other natural and social sciences; Geography as the study of areal differentiation and spatial organisation; Defining the field of human geography, reactions to nomothetic and ideographic approaches	12
Unit 2	Geography through the ages; general character of geographic knowledge during the ancient and mediaeval period; impact of discoveries and European renaissance on the emergence of modern geography, Foundations of modern geography: Contribution of German (Humboldt, Ritter, Ratzel), French (Paul Vidal de la Blache), British and American geographers.	12
Unit 3	Evolution of geographic thought (Determinism- the impact of Darwinism, Possibilism, Human Ecology, Morphology of Landscape), , Areal differentiation) and their impact in the development of the field, Post Structuralism and Post Colonialism	10
Unit 4	Positivism and quantitative revolution, behaviouralism, radicalism, humanism and post-modernism, locational analysis, Explanation in geography: laws and theories; models and system analysis, Spatial analysis: history and concept of space and spatial organisation, gender geography and post modernism geography	14

	Total	**48**

Text Books:

1. Adhikari, S., 1992: Geographical Thought, Chaitanya Pub. House, Allahabad.
2. Berry, B. J. L., 1973: 'A Paradigm for Modern Geography', in R. J. Chorley (ed), Directions in Geography, London Methuen.
3. Bunge, W., 1962: Theoretical Geography, Lund Studies in Geography, Lund, C.W.K. Gleerup.
4. Buttimar, A., 1978: 'On People, Paradigms and Progress in Geography', in D.R. Stoddart (ed),
5. Geography, Ideology and Social Concern, Oxford, Blackwell.
6. Dickinson, R. E., 1969: Makers of Modern Geography, Routledge and Kegan Paul, London.

Reference Books:

1. Dikshit, R. D., 1997: Geographical Thoughts: A Contextual History of Ideas, Prentice Hall of India, New Delhi.
2. Gold, J. R., 1980: An Introduction to Behavioural Geography, Oxford University Press.
3. Hartshorne, R., 1939: The Nature of Geography, Association of American Geographers, Lancaster, Penn.
4. Hartshorne, R., 1959: Perspective on the Nature of Geography, Rand Mckully, Chicago.
5. Harvey, D., 1969: Explanation in Geography, St. Martin's Press, New York.
6. Harvey, Milton and Holly, Brian P.1989: Themes in Geographic Thought, Routledge, London.
7. James, P. E., 1972: ***All Possible World: A History of Geographic Ideas***, The Odyssey Press, New York.

Paper III Core Course	GEOGRAPHY OF RURAL DEVELOPMENT			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO164C203

Course Objectives: *The course aims to make students aware of the concepts, approaches, and planning processes related to rural development in India, along with understanding the rural economic base, rural development strategies, and provision of services in rural areas. It emphasizes sustainable rural transformation, aligning with* ***SDG 1: No Poverty****,* ***SDG 2: Zero Hunger****,* ***SDG 3: Good Health and Well-being****, and* ***SDG 10: Reduced Inequalities****.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO. No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1	Define the need and approaches to rural development, contributing to **SDG 1: No Poverty**.	BT1
CO2	Interpret in detail the rural economic base, especially the significance of developing the non-farm sector in rural areas, aligned with **SDG 8: Decent Work and Economic Growth**.	BT2
CO3	Develop in-depth knowledge of the pre- and post-independence periods of rural development, supporting **SDG 2: Zero Hunger** and **SDG 10: Reduced Inequalities**.	BT3
CO4	Analyze the relevance of access to services like health and education in rural areas, directly addressing **SDG 3: Good Health and Well-being** and **SDG 4: Quality Education**.	BT4
CO5	Interpret various types of rural development data along with critical evaluation, reinforcing **SDG 9: Industry, Innovation and Infrastructure**.	BT5
CO6	Design and develop a detailed project report for rural development planning, advancing **SDG 11: Sustainable Cities and Communities** and **SDG 17: Partnerships for the Goals**.	BT6

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Rural Development: meaning and dimensions; Need for Rural Development studies in geography; Rural Poverty; Rural development in India: Role of Zilla Parishad and Panchayats in Rural Development, Rural Finances – Banks, NABARD etc.	10
Unit 2	Concept of Village, Rural Settlement pattern, Rural Economic Base: Panchayati raj System, Agriculture and Allied Sectors, Seasonality and Need for Expanding Non-Farm Activities, Co-operatives	14
Unit 3	The Problem of Housing, housing types, low-cost houses, the housing schemes in rural area, Dimensions of Rural unemployment and under employment; Rural – Urban migration issues.	10
Unit 4	Recent Rural Development programmes in India: MGNREGA, DAY-NRLM, DDU-GKY, PMAY-G, SPMRM, PMKSY, PM-KUSUM, SVAMITVA Scheme, PM Surya Ghar: Muft Bijli Yojana, Vishwakarma Yojana	14
	Total	**48**

Text Book:

1. Gilg A. W., 1985: An Introduction to Rural Geography, Edwin Arnold, London.
2. Krishnamurthy, J. 2000: Rural Development - Problems and Prospects, Rawat Publs., Jaipur
3. Lee D. A. and Chaudhri D. P. (eds.), 1983: Rural Development and State, Methuen, London.
4. Misra R. P. and Sundaram, K. V. (eds.), 1979: Rural Area Development: Perspectives and Approaches, Sterling, New Delhi.

References:

1. Robb P. (ed.), 1983: Rural South Asia: Linkages, Change and Development, Curzon Press.
2. UNAPDI 1986:Local Level Planning and Rural Development: Alternative Strategies. (United Nations Asian & Pacific Development Institute, Bangkok), Concept Publs. Co., New Delhi.
3. Wanmali S., 1992: Rural Infrastructure Settlement Systems and Development of the Regional Economy in South India, International Food Policy Research Institute, Washington, D.C.
4. Yugandhar, B. N. and Mukherjee, Neela (eds.) 1991: Studies in Village India: Issues in Rural Development, Concept Publs. Co., New Delhi.

Misra, R. P. (ed.), 1985: Rural Development: Capitalist and Socialist Paths, Vol. 1, Concept, New Delhi

Paper IV Core Course	Urban Geography L-T-P-C: 4-0-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEO164D204

Course Objectives: *The course aims to give students a clear understanding of the concept of urban geography and its major aspects, while developing new insights into the relevance of urban geography and its associated challenges in a rapidly urbanizing world. It emphasizes sustainable urban development and resilience, aligning with* ***SDG 11: Sustainable Cities and Communities*** *for inclusive urban planning,* ***SDG 13: Climate Action*** *for addressing urban climate challenges, and* ***SDG 10: Reduced Inequalities*** *for equitable access to urban resources.*

Course outcomes:

After successful completion of the course, the students will be able to:		
CO No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	Define concepts related to urban geography and its approaches, contributing to **SDG 11: Sustainable Cities and Communities**.	**BT 1**
CO2.	Explain different geographical factors that organize urban spaces and develop ideas in relation to them, aligned with **SDG 11: Urban Planning** and **SDG 13: Climate Action**.	**BT 2**
CO3.	Identify new insights on the relevance of urban geography in contemporary times, supporting **SDG 11: Sustainable Cities and Communities** and **SDG 10: Reduced Inequalities**.	**BT 3**
CO4.	Discover and develop skills for advanced studies in urban planning and development, reinforcing **SDG 11: Sustainable Cities and Communities** and **SDG 13: Climate Action**.	**BT 4**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Urban Geography: Meaning, subject matter and scope; approaches and trends in urban geography. Towns: Types, characteristics, origin and growth in global and national contexts, Functional classification of towns; Schemes of city classification	10
Unit 2	Patterns of Urbanization in developed and developing countries; Components of urbanization and urban population growth, Urban morphology and land use structure; Theories on the internal structure of town: the Sector Theory of Homer and Hoyt, and the Multiple Nuclei Theory of Harris and Ullman	14
Unit 3	Concept of city-region, urban agglomeration, urban sprawl, Umland and periphery, rural-urban dichotomy and continuum, urban fringe, satellite town, new town, smart city. Urban Systems: Concept of urban system and hierarchy; Christaller's Central Place Theory; the rank-size distribution of cities; concept of primate city.	14
Unit 4	Urban issues and problems: Housing, slums, civic amenities (transportation and drinking water), traffic congestion, pollution (air, noise, water), and crime. Urbanization and urban development planning in India: Trend and regional patterns of urbanization; national urban development policies and programmes; emerging urban issues of selected cities (Delhi NCR, Mumbai, Guwahati).	10

	Total	48

Text Book:

1. Bansal, S.C. (2010): Urban Geography, Meenakshi Prakashan, Meerut.
2. Hall T., 2006: Urban Geography, Taylor and Francis.
3. Kaplan D. H., Wheeler J. O. and Holloway S. R., 2008: Urban Geography, John Wiley.
4. Knox P. L. and McCarthy L., 2005: Urbanization: An Introduction to Urban Geography, Pearson Prentice Hall New York.
5. Pacione M., 2009: Urban Geography: A Global Perspective, Taylor and Francis.

References:

1. Bala, R. (1986): Urbanisation in India, Rawat, Jaipur.
2. Fyfe N. R. and Kenny J. T., 2005: The Urban Geography Reader, Routledge.
3. Graham S. and Marvin S., 2001: Splintering Urbanism: Networked
4. Infrastructures, Technological Mobilities and the Urban Condition, Routledge
5. Knox P. L. and Pinch S., 2006: Urban Social Geography: An Introduction, Prentice Hall
6. Kundu, A. (1992): Urban Development and Urban Research in India, Khanna Publication, New Delhi.
7. Ramachandran R (1989): Urbanisation and Urban Systems of India, Oxford University Press, New Delhi

Paper V Core Course	Traditional and Digital Cartographic Techniques L-T-P-C: 0-0-8-4 Credit Units: 4 Scheme of Evaluation: (P)	Subject Code: GEO164C215

Course Objectives: *This course aims to highlight the importance of various surveying techniques in geographical study, including concepts of scale and map projections suited for different purposes. It also emphasizes field ethics and the use of traditional and modern tools of field study. The course aligns with* ***SDG 9: Industry, Innovation and Infrastructure*** *for geospatial innovation,* ***SDG 11: Sustainable Cities and Communities*** *for urban and regional planning, and* ***SDG 15: Life on Land*** *for environmental monitoring and conservation.*

Course Outcomes:

By the end of this course the students will be able to:		
CO No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	Recall varied cartographic terms, terminologies, and techniques, contributing to **SDG 4: Quality Education**.	**BT 1**
CO2.	Construct different types of map projections, supporting **SDG 9: Innovation and Infrastructure**.	**BT 2**
CO3.	Develop skills in preparing thematic maps at various levels, aligned with **SDG 11: Sustainable Cities and Communities**.	**BT 3**
CO4.	Analyze GIS-based maps, perform spatial analysis, classify remote sensing satellite data, and prepare large-scale maps using traditional surveying equipment and GPS, reinforcing **SDG 9: Innovation** and **SDG 11: Urban Planning**.	**BT 4**
CO5.	Assess multiple surveying techniques and their applications, directly addressing **SDG 15: Life on Land** and **SDG 11: Sustainable Cities and Communities**.	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Population projection model and graph of India /North East India/ Any State of India - **1 Exercise;** Cartogram for socio-economic or any other data of India or North East India - Urban population by proportionate sphere or circles - **1 Exercise;** Preparation of one quantitative thematic map (Choropleth technique or Isopleth technique) by using district level data of any state of India - **1 Exercise**	12
Unit 2	Map Projection: Basic concepts, classification, basic Principles of construction of zenithal, conical and cylindrical groups of map projections and choice of map projections. Construction of graticules and drawing of maps thereon including properties and uses of: Zenithal Gnomonic Projection (Equatorial case) Mollweide's Projection Conical Projection with two-standard parallels Lambert's Conical Equal Area Projection Mercator's Projection Preparation of topographic maps / profile/ contouring by using: Plane Table / Prismatic Compass Surveying – **(1 Exercise)** Profile / Contouring by Dumpy Level / Autolevel- **(1 Exercise)**	12
Unit 3	Georeferencing of maps by using Open Source GIS / Professional GIS Software **(1 Exercise)**, Map in UTM Projection System **(1 Exercise),** Map digitization and creation of point, line and polygon layers **(3 Exercises),** Preparation of thematic maps **(2 Exercises)**;	12

Unit 4	Digital Image Processing- Unsupervised Classification and Supervised Classification **(2 Exercises)**; Geographic Analysis of Spatial data: Spatial Mean Centre and Standard Distance **(1 Exercise)**; Preparation of maps by using GPS data **(1 Exercise),** Digital Elevation Modelling creation through different techniques**(2 exercise)**	12
	Total	**48**

Text Books:

As per the list of GEO164C203 and GEO164C204

Reference Books:

As per the list of GEO164C203 and GEO164C204

SCHEME OF EVALUATION (GEO164C215)

- Internal Assessment and Mid-term Examinations: As per the University rules
- End Semester Examination should consist of the questions taking any one exercise from each module. The module-III and IV should have optional questions.
- The distribution of marks for each module will remain equal as far as practicable (Each question with10% -20% variation in the distribution of marks may be set based on difficulty level and time requirements to perform the exercise). 80% of the total marks is reserved for 4 exercises to be performed with total duration of 4 (four) hours
- 10% of total marks is reserved for External Examiner
- 10% of total marks is reserved for Practical Records and Note Book
- Maintenance of Practical Records and Note Book is mandatory
- External Examiner is mandatory during End Semester Examination

SEMESTER III

Paper I Core Course	REMOTE SENSING, GIS AND GPS: PRINCIPLES AND APPLICATIONS			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO164C301

Course Objectives: *The course aims to make students interpret data, tools, and technology in Geoinformatics — GIS, Remote Sensing, and GPS — and construct and analyse maps using geospatial technology. It emphasizes the application of geoinformatics for sustainable development, aligning with* ***SDG 9: Industry, Innovation and Infrastructure*** *for technological innovation,* ***SDG 11: Sustainable Cities and Communities*** *for urban and regional planning, and* ***SDG 13: Climate Action*** *for climate resilience and adaptation.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
CO No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	Define the fundamental terms and terminologies of Geoinformatics, contributing to **SDG 4: Quality Education**.	**BT 1**
CO2.	Outline the strength and applications of geospatial technology, supporting **SDG 9: Innovation and Infrastructure**.	**BT 2**
CO3.	Build maps of resources, their location, and availability, aligned with **SDG 11: Sustainable Cities and Communities**.	**BT 3**
CO4.	Analyse different remote sensing datasets collected from various platforms, reinforcing **SDG 13: Climate Action** and **SDG 15: Life on Land**.	**BT 4**
CO5.	Interpret geospatial data in GIS platforms and perform analysis using data from Remote Sensing and GPS for geographical research, directly addressing **SDG 9: Innovation**, **SDG 11: Urban Planning**, and **SDG 13: Climate Action**.	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Needs and applications of Aerial photography; Aerial photography platforms; Planning aerial photography survey missions; Aerial camera and film characteristics; Types of aerial photographs. Earth observation satellites (EOS) and Remote Sensing (RS) satellites - orbital characteristics; Payloads of RS satellites; Types and characteristics of sensors; Spatial, radiometric, spectral and temporal resolutions of RS data. Electromagnetic radiation (EMR) principles.	12
Unit 2	Geographical Information Sciences – a brief history; Definitions of GIS, Components of a GIS; Referencing systems: International Terrestrial Reference Frame (ITRF); Ellipsoids and datum, Coordinate Systems; GIS data types and structure, Types of GIS Software Systems; Database Management Systems; Digital Elevation Model (DEM) and its derivatives; Triangulated Irregular Network (TIN) data structure and its characteristics. Geoprocessing - Buffering and Proximity analysis	12
Unit 3	Basic concepts of GPS, Working principles of GPS, GPS errors and their corrections, GPS Receivers and their characteristics, Differential Positioning, Accuracy in GPS and DGPS, GPS surveying methods; Topographic mapping and GIS applications for GPS / DGPS.	10
Unit 4	Image classification techniques (Supervised and Unsupervised); Major classifiers / classification functions; Change detection analysis and accuracy assessment methods. Basic ideas and applications of thermal, hyperspectral, microwave and LiDAR Remote Sensing. Applications of Remote Sensing, GIS and GPS: Urban planning and management, Environmental monitoring and management, Natural resource management, Disaster management.	14
	Total	**48**

Text Books:

1. Burrough, P. A.,1986: Principles of Geographical Information Systems in Land Resources Assessment, Clarendon Press, Oxford
2. Burrough, P. A. and McDonnell, R. A., 1998: Principles of Geographical Information Systems, Oxford University Press, Oxford.
3. Curtis, H., 2000: The GPS Accuracy Improvement Initiative, GPS World, June, 2000.
4. Chrisman, N., 1997: Exploring Geographic Information Systems, John Wiley & Sons Inc.,
5. De Mars, M. N., 1999: Fundamentals of Geographic Information Systems, John Wiley & Sons Inc., New York.
6. Gopi, S., 2005: Global Positioning System Principles and Applications, Tata McGraw Hill, New Delhi.
7. Kraaak, M. and Ormelling, F., 2004: Cartography Visualization of Geospatial Data, Pearson Education, Delhi.

Reference Books:

1. Chetry, N., (Editor) 2019: A Glimpse of Geospatial Technologies and Applications, EBH Publishers (India), Guwahati
2. Star, J. and Ester, J., 1990: Geographic Information System, Prentice-Hall.
3. Maguire, D. J., Goodchild, M. and Rhind, D. J., 1990: Geographical Information Systems: Principles and Applications, Longman Science and Technology Publications.
4. Robinson, A. H., et al., 1995: Elements of Cartography, John Wiley.

CORE COURSE II	SOCIAL AND CULTURAL GEOGRAPHY L-T-P-C: 4-0-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Course code GEO164C302

Course Objectives: *The course aims to make students understand the basic concepts related to social and cultural geography in the geographical framework in the spatial context. The course also aims to promote awareness of global development concerns, in relation to SDG 10: Reduced Inequalities, by encouraging students to critically examine issues of urban sustainability, cultural diversity, social inclusion, and spatial inequalities within human societies.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
SI No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	**Define** the fundamental concepts of social and cultural dimensions.	**BT 1**
CO2.	**Interpret** the social and cultural concepts in a broader and analytical manner	**BT 2**
CO3.	**Build** knowledge on structures, formations of countries as well as on various schools of social and cultural geography	**BT 3**
CO4.	**Discover** the humanistic perspective and its dimensions in Geography in relation to the physical and cultural surrounding fostering the ideals of SDG 10: Reduced Inequalities through a deeper understanding of human experiences and inequalities across different communities and regions.	**BT 4**
CO5.	**Perceive the** issues related to socio-cultural aspects with a geographical perspective.	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Meaning, scope and subject matter of Social Geography, Growth and development of Social Geography, Meaning and characteristics of Society, Types of Society: Tribal, agrarian and industrial, Social Groups: meaning, classification and characteristics, Primary and secondary groups, Concept of social space	12
Unit 2	Social process: Social Conflict and cooperation: meaning, characteristics and forms; social justice, Social well-being: meaning and indicators , Spatial distribution of social groups in India	12
Unit 3	Meaning, scope and development of cultural geography, Concept of culture, characteristics of culture, its components and functions, cultural region: meaning and types. Cultural regions of the world.	10
Unit 4	Concept of cultural landscape, cultural integration, cultural diffusion, cultural heritage, cultural ecology, cultural convergence, cultural lag. Language and religion: worldwide distribution, Cultural diversity in North-east India and Assam	14
	Total	**48**

Text Books:

1. Sen, J (2016): A text book of Social and Cultural Geography
2. Dwiveda R. L. : Fundamentals of Political Geography
3. Pounds N .J. G. (1972): Political Geography, McGraw Hill, New York.

Reference Books:

1. Jordan, T. G. and Rowntree, L.: The Human Mosaic: A Thematic Interpretation in Cultural Geography
2. John R. S., 1982: An introduction to Political Geography, Routledge, London.
3. Ahmad, A., 1999: Social Geography, Rawat Publication, Jaipur and New Delhi.

4. Ahmad, A. (ed), 1993: Social Structure and Regional Development: A Social Geography perspective, Rawat Publication, Jaipur.
5. Noble, A. G. and Dutta, A. K. (eds): India: Cultural Pattern and Processes, West View Press, Colorado.

Paper III Core Course	RESEARCH METHODOLOGY IN GEOGRAPHY			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO164D303

Course Objectives: *The course aims to make students understand the basics of qualitative and quantitative research, literature review, data collection, identification of research problems, formulation of research objectives and questions, hypothesis development and testing, framing of questionnaires, techniques of collecting both qualitative and quantitative data, and their analysis. It emphasizes research for sustainable development, aligning with* ***SDG 4: Quality Education*** *for skill development,* ***SDG 9: Industry, Innovation and Infrastructure*** *for innovation through research, and* ***SDG 17: Partnerships for the Goals*** *for collaborative knowledge creation.*

Course Outcomes:

After the completion of course, the students will have ability to:		
CO No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	Define the concepts and tools of research, contributing to **SDG 4: Quality Education**.	**BT 1**
CO2.	Infer ideas for research work through literature review, aligned with **SDG 4: Quality Education** and **SDG 9: Innovation**.	**BT 2**
CO3.	Develop hypotheses and research questions, reinforcing **SDG 9: Industry, Innovation and Infrastructure**.	**BT 3**
CO4.	Identify appropriate data collection and sampling techniques, supporting **SDG 4: Quality Education**.	**BT 3**
CO5.	Interpret various types of data along with critical evaluation, directly addressing **SDG 9: Innovation**.	**BT 5**
CO6.	Design and develop a scientific research report, advancing **SDG 17: Partnerships for the Goals** through collaborative and applied research.	**BT 6**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Research: Definition, types, significance and important features; Research methodology in geography; Defining a research problem; Statement of the problem; Objectives, and hypothesis/ research questions, Database and methodology, significance, review of research works and bibliography and references.	12
Unit 2	Research design: Meaning, need and features of a good design, Inductive and deductive approaches in geographic research, Hypothesis, theories and models in Geography; concept of model building and hypothesis testing, Research ethics: Plagiarism-classification and prevention; Intellectual property rights; Research report: Structural components and presentation.	12

Unit 3	Field Work in Geographical Studies – Role, Value, Data and Ethics of Field-Work; Defining the Field and Identifying the Case Study – Rural / Urban / Physical / Human / Environmental; Designing the Field Report – Aims and Objectives, Methodology, Analysis, Interpretation and Writing the Report	12
Unit 4	Sources of geographic data (Conventional and Geospatial technology based), their representation, interpretation and analysis; Field Tools and Techniques – Merits, Demerits and Selection of the Appropriate Technique; Observation (Participant / Non Participant), Questionnaires (Open/ Closed / Structured / Non-Structured) Interview with Special Focus on Focused Group Discussions	12
	Total	**48**

Note: Computer / calculator based compulsory home assignments may be given for various units. Scientific calculator may be permitted in the examination hall for this paper.

Text Books:

1. Harvey, D, 1969: Explanation in Geography, Scientific Publisher, Jodhpur.
2. Lenon, B., Cleves, P. 2015. Geography Fieldwork and Skills, Harper-Collins.
3. Montello , D.R, Sutton, P. 2012. An Introduction to Scientific Research Methods in Geography and Environmental Studies, 2nd ed, Sage.
4. Murthy , K.L.N. 2004. Research Methodology in Geography: A Text Book, Concept Publishing Co.

Reference Books:

1. Evans, M., (1988): "Participant Observation: The Researcher as Research Tool" in Qualitative Methods in Human Geography, eds. J. Eyles and D. Smith, Polity.
2. Special Issue on "Doing Fieldwork" The Geographical Review 91:1-2 (2001).
3. Stoddard, R. H., (1982): Field Techniques and Research Methods in Geography, Kendall/Hunt.
4. Wolcott, H., (1995): The Art of Fieldwork, Alta Mira Press, Walnut Creek, CA.
5. Northey, N., Draper, D., Knight, D.B. 2015. Making Sense in Geography and Environmental Sciences:A Student's Guide to Research and Writing, 6th ed, Oxford University Press.
6. Parsons, T., Knight, P.G. 2015. How To Do Your Dissertation in Geography and Related Disciplines,3rd ed, Routledge.

Paper IV Core Course	DISASTER MANAGEMENT			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO164C304

Course Objectives: *The objective of this course is to make students understand hazards and disasters, their associated causes and impacts, distribution, and mitigation with special reference to India. It emphasizes disaster resilience and preparedness, aligning with* ***SDG 3: Good Health and Well-being*** *for reducing disaster-related health risks,* ***SDG 6: Clean Water and Sanitation*** *for safeguarding water resources during disasters, and* ***SDG 13: Climate Action*** *for climate resilience and adaptation.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
SI No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	Define the concepts of hazard and disaster and related terminologies, contributing to **SDG 13: Climate Action**.	**BT 1**
CO2.	Demonstrate the distribution and mapping of disasters prevalent in India, aligned with **SDG 11: Sustainable Cities and Communities** and **SDG 6: Clean Water and Sanitation**.	**BT 2**
CO3.	Explain the mitigation processes and responses to disasters across Indian territory, supporting **SDG 3: Good Health and Well-being** and **SDG 13: Climate Action**.	**BT 2**
CO4.	Distinguish between causes and effects of varied disasters, as well as their implications in present-day India, reinforcing **SDG 11: Sustainable Cities and Communities** and **SDG 13: Climate Action**.	**BT 4**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Disasters: Definition and Concepts: Hazards, Disasters; Risk and Vulnerability; Classification	12
Unit 2	Disasters in India: (a) Flood: Causes, Impact, Distribution and Mapping; Landslide: Causes, Impact, Distribution and Mapping; Drought: Causes, Impact, Distribution and Mapping	12
Unit 3	Disasters in India: (b) Earthquake and Tsunami: Causes, Impact, Distribution and Mapping; Cyclone: Causes, Impact, Distribution and Mapping. Manmade disasters: Causes, Impact, Distribution and Mapping	12
Unit 4	Response and Mitigation to Disasters: Mitigation and Preparedness, NDMA and NIDM; Indigenous Knowledge and Community-Based Disaster Management; Do's and Don'ts During and Post Disasters	12
	Total	**48**

Text Book:

1. Singh, R. B. (ed.), (2006) Natural Hazards and Disaster Management: Vulnerability and Mitigation, Rawat Publications, New Delhi.
2. Sinha, A. (2001). Disaster Management: Lessons Drawn and Strategies for Future, New United Press, New Delhi.

References:

1. Government of India. (1997) Vulnerability Atlas of India. New Delhi, Building Materials & Technology Promotion Council, Ministry of Urban Development, Government of India.
2. Kapur, A. (2010) Vulnerable India: A Geographical Study of Disasters, Sage Publication, New Delhi.
3. Modh, S. (2010) Managing Natural Disaster: Hydrological, Marine and Geological Disasters, Macmillan, Delhi.
4. Singh, R.B. (2005) Risk Assessment and Vulnerability Analysis, IGNOU, New Delhi.
5. Stoltman, J.P. et al. (2004) International Perspectives on Natural Disasters, Kluwer Academic Publications. Dordrecht.
6. Singh Jagbir (2007) "Disaster Management Future Challenges and Opportunities", 2007. Publisher- I.K. International Pvt. Ltd. S-25, Green Park Extension, Uphaar Cinema Market, New Delhi, India (www.ikbooks.com).

Paper V Core Course	**Report Writing on Environmental Issues** **Credit Units: 4 Scheme of Evaluation: (P)**	**Subject Code:** **GEO164C325**

Course Objectives: This course intends to make students understand the various dimensions of field work and its role in geographical studies, along with introducing basic report writing and field tools. It emphasizes experiential learning and practical application, aligning with **SDG 4: Quality Education** for skill development, **SDG 11: Sustainable Cities and Communities** for applying field insights to urban and regional planning, and **SDG 15: Life on Land** for ecosystem monitoring and conservation.

Course Outcomes:

After successful completion of the course, the students will be able to:		
Sl. No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1	Choose new geographical landscapes as study areas, contributing to **SDG 15: Life on Land**.	BT1
CO2	Infer in-depth knowledge of different field techniques, aligned with **SDG 4: Quality Education**.	BT2
CO3	Choose appropriate field ethics and tools of field study, supporting **SDG 11: Sustainable Cities and Communities**.	BT3
CO4	Analyze different field techniques in detail, reinforcing **SDG 9: Industry, Innovation and Infrastructure** and **SDG 15: Ecosystem Conservation**.	BT4

Detailed Syllabus:

Modules	**Topics and Course Content**	**Periods**
Unit 1	Selection of a nearby urban, industrial, or agricultural area facing notable environmental degradation (e.g., pollution, soil erosion, water scarcity, waste mismanagement), Field reconnaissance survey: preparation of base maps using GPS, Google Earth, or topographic sheets, Mapping of physical and environmental features: land use/land cover patterns, pollution sources (air, water, soil, noise), waste disposal sites, vegetation status, etc, Observation and documentation of visible environmental problems: solid waste accumulation, sewage disposal, chemical runoff, loss of biodiversity, etc, Interaction with local stakeholders for preliminary insights into the problem, Maintenance of field diaries and photographic documentation.	12

Unit 2	Framing the core environmental problem for research based on field observations, Compilation of academic and policy literature from journals, books, government reports, and case studies related to the identified problem, Critical review of existing studies and theoretical frameworks (e.g., environmental degradation, pollution control, sustainability, urban ecology), Identification of research gaps and formulation of specific research questions/hypotheses, Development of conceptual models or problem trees to understand causal relationships.	12
Unit 3	Preparation of structured and semi-structured questionnaires for stakeholders (residents, farmers, local authorities), Surveys, interviews, transect walks, environmental sampling (air, water, soil, etc.), Recording of qualitative and quantitative data, Use of census reports, municipal records, pollution control board data, weather records, satellite images, etc., Data coding, entry, and basic statistical summarization using Excel or any GIS/statistical software	12
Unit 4	Generation of report (within about 100 A4 size pages including 30-40 maps/diagrams/field photographs) on the basis of field works carried out under , Final project presentation by each student using PowerPoint during on the scheduled date of viva-voce examination of this paper.	12
	Total	**24**

Text Books:

1. Creswell, J., (1994). ***Research Design: Qualitative and Quantitative Approaches***. UK: Sage Publications.
2. Dikshit, R. D. (2003). ***The Art and Science of Geography: Integrated Readings***.

Reference books:

1. Evans, M. (1988). Participant Observation: The Researcher as Research Tool. In Eylesand, J and D. Smith (eds). Qualitative Methods in Human Geography. Cambridge, UK: Polity.
2. Mukherjee, N. (2002). Participatory Learning and Action: with 100 Field Methods. Delhi, India: Concept Publs. Co.
3. Stoddard, R. H. (1982). Field Techniques and Research Methods in Geography. USA: Kendall/Hunt.

Wolcott, H. (1995). The Art of Fieldwork. CA, USA: Alta Mira Press.

Paper IV Core Course	SURVEY/RESEARCH I L-T-P-C: Credit Units: 8 Scheme of Evaluation: (P)	Subject Code: GEO164C324

Learning Objectives*: This paper provides an understanding of the basics of research project preparation, including problem identification, literature review, hypothesis formulation, data collection, analysis, and report writing. It emphasizes research as a tool for sustainable development, aligning with* ***SDG 4: Quality Education*** *for skill enhancement,* ***SDG 9: Industry, Innovation and Infrastructure*** *for innovation through applied research, and* ***SDG 17: Partnerships for the Goals*** *for collaborative knowledge creation.*

Learning Outcomes:

After the completion of course, the students will have ability to:		
CO No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1	Relate real-world issues for carrying out research in specific fields, contributing to **SDG 11: Sustainable Cities and Communities**.	**BT 1**
CO2	Infer ideas for research through literature review, aligned with **SDG 4: Quality Education**.	**BT 2**
CO3	Develop hypotheses and research questions, reinforcing **SDG 9: Innovation and Infrastructure**.	**BT 3**
CO4	Identify appropriate sampling techniques, supporting **SDG 4: Quality Education**.	**BT 3**
CO5	Interpret various types of data along with critical evaluation, directly addressing **SDG 9: Innovation**.	**BT 5**
CO6	Design and develop a detailed research report, advancing **SDG 17: Partnerships for the Goals** through collaborative and applied research.	**BT 6**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Identification of Research Problem and Study Area Identification and selection of a geographical research problem based on field observations, internship experiences, or local/regional issues. The study may focus on: a. Agriculture, Industry and Mining b. Environmental Problems, Disasters and Hazards c. Natural Resource Assessment and Management d. Climate Change and Sustainability e. Rural and Urban Development f. Watershed, Basin and Terrain Studies Selection and delineation of study area, statement of the problem, significance and scope of the study.	**12**
Unit 2	Review of Literature and Research Design • Review of relevant literature • Identification of research gaps • Formulation of aims and objectives • Development of research questions and hypotheses • Conceptual and theoretical framework • Preparation of work plan and research design	**12**
Unit 3	Field Survey Design and Methodology • Selection of data sources (Primary and Secondary) • Sampling techniques and survey design	**12**

	• Preparation of questionnaire, schedule and interview formats • GPS-based field data collection techniques • Observation and Participatory Rural Appraisal (PRA) methods • Pilot survey and validation of research tools • Ethical considerations in field research • The field survey is scheduled to commence in the last week of October.	
Unit 4	Research Proposal Preparation and Presentation Preparation of research proposal including: • Title of the Study • Introduction and Background • Statement of the Problem • Objectives and Research Questions • Study Area Description • Database and Methodology • Literature Review • Expected Outcomes • Time Schedule and Work Plan Presentation of the proposal through PowerPoint and discussion before the departmental committee.	**12**
	Total	**48**

Text Books:

As per the list of given in the syllabus for other papers

Reference Books:

As per the list of given in the syllabus for other papers

SEMESTER IV

Paper I Core Course	GEOGRAPHY OF POPULATION AND SETTLEMENT			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO164C401

Course Objectives:

The course aims to make students understand different concepts related to population and their characteristics, including demographic parameters, settlement structures, and the impact of population dynamics on society and the environment. It emphasizes sustainable population management and settlement planning, aligning with ***SDG 3: Good Health and Well-being*** *for demographic health,* ***SDG 10: Reduced Inequalities*** *for equitable population policies, and* ***SDG 11: Sustainable Cities and Communities*** *for sustainable settlement planning.*

Course Outcomes:

By the end of this course the students will be able to:		
CO No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	Tell about the basic definitions and concepts related to population geography and human settlements, contributing to **SDG 4: Quality Education**.	**BT 1**
CO2.	Outline the population parameters of India, aligned with **SDG 3: Good Health and Well-being**.	**BT 2**
CO3.	Apply and analyse the resultant impact of contemporary population issues on society and environment, supporting **SDG 10: Reduced Inequalities** and **SDG 13: Climate Action**.	**BT 3**
CO4.	Analyse contemporary issues related to population dynamics and environment, reinforcing **SDG 11: Sustainable Cities and Communities.**	**BT 4**
CO5.	Determine clear exposition of spatial and structural characteristics of human settlements, directly addressing **SDG 11: Sustainable Cities and Communities**.	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Field of Population Geography, Basic concepts of Population Geography: under population, optimum population, over population, population explosion and population pressure, Population resource relationship, population-resource regions, World population distribution and growth, components of population growth: fertility, mortality, migration.	10
Unit 2	Population composition: age, sex, child women ratio and other related factors, Demographic transition model and theories of population Growth (Malthus, Sadler and Ricardo), Population policies in developed and developing economies; Gender and Space: Safety, Mobility, Exclusion; Spatial patterns of gender disparities in sex ratio differentials, female literacy, work force participation; gender, health and access to healthcare; land ownership and property rights; patterns of participation in local and national politics.	14
Unit 3	Defining the field and scope of Settlement Geography and its approaches, origin and growth of settlement: rural and urban, types and pattern of settlement, concepts of rural-urban dichotomy, continuum and urban fringe, urban system (Law of primate city and rank size rule).	10
Unit 4	Concept of urbanization and westernisation, functional classification of urban settlements, internal structure of cities / settlement hierarchy with reference to central place theory (Christaller and Losch): Measurement of centrality, nodality and hierarchy	14
	Total	**48**

Text Books:

1. Singh R.Y. (Rep. 2010) - Geography of Settlements, ShardaPustakBhawan, Allahabad
2. Chandna R. C. (Rep.2010) – A Geography of Population, Concepts, Determinants andPatterns, Kalyani Publishers, New Delhi.
3. Maurya S.D (Rep. 2018): Settlement Geography, ShardaPustakBhawan, Allahabad
4. Sandram, K. V. and Nangia, S., (eds): Population Geography, Heritage Publishers, New Delhi. Inc., New York.
5. Ghosh, S. (2015) Introduction to Settlement Geography, Orient Black Swan Private Ltd., Kolkata.

Reference Books:

1. Clarke, J. I., 1972: Population Geography,Pergamon Press, Oxford.
2. Peters, G. L. and Larkin, R. P., 1979: Population Geography: Problems, Concepts and Prospects,Kendall/ Hunt Iowa.
3. Trewartha, G. T., 1969: A Geography of Population: World Pattern, John Wiley & Sons.
4. Woods, R., 1979: Population Analysis in Geography, Longman, London.
5. Robinson, H., 1981: Population and Resources, Macmillan Press, London
6. Kaushik, S.D. (2010) ManavBhugol, Rastogi Publication, Meerut.
7. Maurya, S.D. (2012) ManavBhugol, ShardaPustakBhawan, Allahabad.

Core course II	Political Geography			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO164C402

Course Objectives: This course aims to introduce students to the theoretical dimensions of political geography, focusing on the political organization of space. The course promotes awareness of **SDG 16: Peace, Justice and Strong Institutions** (sub-themes: 16.6 Governance, 16.7 Inclusive institutions), encouraging students to critically assess how demographic dynamics and geopolitical processes intersect with issues of equity, governance, and inclusive political participation.

Course Outcomes:

By the end of this course the students will be able to:		
CO.	**Course Outcome**	**Blooms Taxonomy Level**
CO1	**Define** the parameters of political geography and relate them to principles of social inclusion	BT1
CO2	**Interpret** spatial patterns of population and political boundaries in the context of equitable governance (SDG 16.6).	BT2
CO3	**Identify** political geographic concepts such as state, territory, and boundaries in relation to inclusive institutions and democratic participation (SDG 16.7).	BT3
CO4	**Analyse** how population dynamics and political processes reinforce or reduce inequalities across regions, aligned with SDG 16.	BT4
CO5	**Appraise** the socio-cultural and political theme in the geographical dimensions	BT5

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	**Foundations of Political Geography and Geopolitics**: Nature, scope, evolution and contemporary approaches of Political Geography; relationship between Political Geography and Geopolitics; major themes and methodological approaches in Political Geography; historical development of Political Geography and major schools of thought. Concepts, scope and significance of Geopolitics in contemporary world affairs.	12
Unit 2	**Elements and Concepts in Political Geography**: Physical, economic, demographic and socio-cultural elements in Political Geography; concepts of core area, ecumene, capital city and political integration; frontiers and boundaries: meaning, types and functions; concepts of Nation, State and Nation-State; concept and geopolitical significance of buffer states; interstate boundary disputes in North-East India with special reference to territorial and ethnic dimensions.	12
Unit 3	**The State and Political Organization of Space**: Concept, origin, evolution and classification of the State; attributes of the State—territory, population, sovereignty and government; theories related to the rise, growth and disintegration of states; political evolution and territorial integration of India as a modern state; dynamics of nationalism, regionalism and federalism in India.	10
Unit 4	**Geostrategic Theories and International Boundaries**: Classical and contemporary geopolitical theories with special reference to Heartland Theory and Rimland Theory; international boundaries: nature, functions and morphological classification; maritime boundaries, territorial waters, Exclusive Economic Zone (EEZ) and Law of the Sea; role of the United Nations in codifying maritime laws; major international boundary disputes with special reference to India–China and India–Pakistan relations and border conflicts	14
	Total	**48**

Text Books:

1. Dwivedi, R L., 2014: Fundamentals of Political Geography, Allahabad.
2. Pounds N .J. G. (1972): Political Geography, McGraw Hill, New York.

Reference Books:

1. John R. S., 1982: An introduction to Political Geography, Routledge, London.
2. Ahmad, A., 1999: Social Geography, Rawat Publication, Jaipur and New Delhi.

3. Ahmad, A. (ed), 1993: Social Structure and Regional Development: A Social Geography perspective, Rawat Publication, Jaipur.

Paper III Core Course	Biodiversity Conservation			Subject Code:
	L-T-P-C: 2-0-4-2	Credit Units: 2	Scheme of Evaluation: (T+P)	GEO164C403

Course Objectives: *This course intends to make students understand the importance of biodiversity and its conservation. It emphasizes ecological balance, sustainable resource use, and conservation strategies, aligning with* ***SDG 15: Life on Land*** *for protecting ecosystems,* ***SDG 13: Climate Action*** *for climate resilience, and* ***SDG 14: Life Below Water*** *for marine biodiversity protection*.

Course Outcomes:

After the completion of the course, the students will have the ability to:		
Sl. No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1	Build ideas on the importance of biodiversity, contributing to **SDG 15: Life on Land**.	BT1
CO2	Analyse the ways of biodiversity conservation at micro, meso, and macro levels, aligned with **SDG 13: Climate Action** and **SDG 14: Life Below Water**.	BT2
CO3	Interpret various techniques of biodiversity conservation, reinforcing **SDG 15: Ecosystem Protection** and **SDG 12: Responsible Consumption and Production**.	BT3

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Threats: Factors causing biodiversity degradation, Concept of species extinction. Traditional Conservation Practice: Traditional Ecological Knowledge and documentation of local biodiversity.	6
Unit 2	Species Conservation Techniques: In situ conservation (Biosphere Reserves, National Parks, Wildlife Sanctuaries, Conservation Reserves, Community reserves, Sacred Habitats),	6
Unit 3	Ex-situ conservation (Botanical & Zoological Gardens, Gene Banks, Seed And Seedling Banks, Pollen Culture, Tissue Culture and DNA banks, Butterfly Gardening); Concept of Biodiversity Hotspots and Mega-diversity Country; Role of captivity in wildlife management.	
Unit 4	Habitat Conservation Techniques: Concept of Habitat, Habitat Management, Habitat Edge improvement, Role of Corridor in Wildlife Management, Ecological Restoration Programme; Social Forestry; Agro Forestry; Joint Forest Management	
	Total	**24**

Text Books:

3. Creswell, J., (1994). ***Research Design: Qualitative and Quantitative Approaches***. UK: Sage Publications.
4. Dikshit, R. D. (2003). ***The Art and Science of Geography: Integrated Readings***.

Reference books:

4. Evans, M. (1988). Participant Observation: The Researcher as Research Tool. In Eylesand, J and D. Smith (eds). Qualitative Methods in Human Geography. Cambridge, UK: Polity.
5. Mukherjee, N. (2002). Participatory Learning and Action: with 100 Field Methods. Delhi, India: Concept Publs. Co.
6. Stoddard, R. H. (1982). Field Techniques and Research Methods in Geography. USA: Kendall/Hunt.
7. Wolcott, H. (1995). The Art of Fieldwork. CA, USA: Alta Mira Press.

Paper IV Core Course	GEOGRAPHY OF TOURISM			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEO164C404

Course Objectives:

*The course aims to give students an understanding of the basic themes and concepts of tourism geography and interpret the geographical components of tourism. It emphasizes sustainable tourism development, aligning with **SDG 8: Decent Work and Economic Growth** for employment and livelihood opportunities, **SDG 11: Sustainable Cities and Communities** for responsible urban and regional tourism planning, and **SDG 12: Responsible Consumption and Production** for sustainable tourism practices.*

Course Outcomes:

By the end of this course the students will be able to:		
SI No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1.	Define the geographical aspects that organize economic space, contributing to **SDG 8: Decent Work and Economic Growth**.	**BT 1**
CO2.	Illustrate the geographical aspects of tourism in an area, aligned with **SDG 11: Sustainable Cities and Communities**.	**BT 2**
CO3.	Develop practical field knowledge about tourist places across India, supporting **SDG 8: Tourism-based Livelihoods** and **SDG 12: Sustainable Tourism Practices**.	**BT 3**
CO4.	Analyse knowledge gathered through field visits and prepare respective reports, reinforcing **SDG 11: Sustainable Cities and Communities**.	**BT 4**
CO5.	Justify the methods of tourist flow in relation to the geographical setting of an area, directly addressing **SDG 12: Responsible Consumption and Production** and **SDG 8: Sustainable Economic Growth**.	**BT 5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Meaning, Scope and contents of Geography of tourism: Definition of tourism and its scope; Factors influencing tourism: historical, natural, socio cultural and economic, Importance of geography in tourism, Types of tourism, Constituents of tourism industry	10
Unit 2	Concept of Sustainable tourism: Definition and principles of sustainable tourism, Dimensions of sustainability, Impact of tourism - Environmental, Social, Cultural, and Economic; Concept of Responsible tourism, Global environmental change and tourism, concept of carrying capacity	14
Unit 3	Resources of Tourism: Destination and resource factors, Mass tourism vs. alternative tourism, Ecotourism, Spatial pattern of Tourism Resources in India-National Parks, Wild life sanctuaries, Tiger Reserves, Biosphere reserves & wetlands., history and culture	10
Unit 4	Understanding tourism system: Leiper's Model, Tourist map design, Travel Flows: models, trends and types, Important tourist destination of India, Tourist destinations in North America, South America, Europe, Africa, Antarctica, Australia and Asia, latitude, longitude and international date line	14
	Total	**48**

Text Books:

1. Bhatia, A. K., 1996: Tourism Development: Principles and Practices, Sterling Publishers, New Delhi.
2. Milton, D., 1993: Geography of World Tourism Prentice. Hall, New York.
3. Sharma J. K. (ed.), 2000: Tourism Planning and Development - A new perspective, Kanishka Publishers, New Delhi.

Reference Books:

1. Robinson, H., 1996: A Geography of Tourism. Macdonald and Evans, London, 1996.
2. Williams Stephen, 1998: Tourism Geography, Routledge, Contemporary Human Geography Series, London.
3. Shaw G. and Williams A. M., 1994: Critical issues in Tourism-A Geographical Perspective, Oxford: Blackwell.

Core Course	Report Writing on Socio-economic Issues L-T-P-C: 0-0-8-4 Credit Units: 4 Scheme of Evaluation: (P)	Subject Code: GEO162C425

Course Objectives:

This paper provides an understanding of the basics of research project preparation, including problem identification, literature review, hypothesis formulation, data collection, sampling, analysis, and report writing. It emphasizes research as a driver of sustainable development, aligning with ***SDG 4: Quality Education*** *for skill enhancement,* ***SDG 9: Industry, Innovation and Infrastructure*** *for innovation through applied research, and* ***SDG 17: Partnerships for the Goals*** *for collaborative knowledge creation.*

Course Outcomes:

After successful completion of the course, the students will be able to:		
Sl. No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1	Relate real-world issues for carrying out research in a specific field, contributing to **SDG 11: Sustainable Cities and Communities**.	BT1
CO2	Infer ideas for research through literature review, aligned with **SDG 4: Quality Education**.	BT2
CO3	Develop hypotheses and research questions, reinforcing **SDG 9: Innovation and Infrastructure**.	BT3
CO4	Identify appropriate sampling techniques, supporting **SDG 4: Quality Education**.	BT4
CO5	Interpret various types of data along with critical evaluation, directly addressing **SDG 9: Innovation**.	BT5
CO6	Design and develop a detailed project report, advancing **SDG 17: Partnerships for the Goals** through collaborative and applied research.	BT6

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Identification of research problem / topic on any one of the following aspects (preferably of local area / District / State) Any kind of geographical studies on socio-economic / cultural issues / demographic problems and characteristics The topic selection / modification may be done just before the 2nd Semester End Examination so that the data collection can be done during semester break.	12
Unit 2	Preparation of Literature Review from the identified problem of research	12
Unit 3	Preparation of questionnaire, Field visit, Primary data collection, Secondary data review.	12
Unit 4	Generation of report (within about 100 A4 size pages including 30-40 maps/diagrams/field photographs) on the basis of field works carried out under , Final project presentation by each student using PowerPoint during on the scheduled date of viva-voce examination of this paper.	12
	Total	**48**

Note: Students will work as an intern during the semester break for 4 weeks after 4th semester. Students not being able to obtain any internship will be assigned project work from the department.

Text Books:

As per the list of given in the syllabus for other papers

Reference Books:

As per the list of given in the syllabus for other papers

Core Course	SURVEY/RESEARCH II L-T-P-C: Credit Units: 12 Scheme of Evaluation: (P)	Subject Code: GEO164C426

Learning Objectives:

The course aims to make students understand how to approach a research problem, formulate research objectives and questions in proper perspective, develop and test hypotheses, frame questionnaires, and apply techniques for collecting both qualitative and quantitative data with appropriate analysis. It emphasizes research as a foundation for sustainable solutions, aligning with ***SDG 4: Quality Education*** *for skill development,* ***SDG 9: Industry, Innovation and Infrastructure*** *for innovation through applied research, and* ***SDG 17: Partnerships for the Goals*** *for collaborative knowledge creation.*

Learning Outcomes:

After the completion of course, the students will have ability to:		
CO No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1	Choose appropriate research methods for carrying out research in a specific field, contributing to **SDG 4: Quality Education**.	**BT 1**
CO2	Infer ideas and classify issues that need attention for hypothesis formulation, aligned with **SDG 9: Innovation and Infrastructure**.	**BT 2**
CO3	Develop hypotheses and research questions, reinforcing **SDG 9: Innovation**.	**BT 3**
CO4	Identify appropriate sampling techniques, supporting **SDG 4: Quality Education**.	**BT 3**
CO5	Interpret various types of data along with critical evaluation, directly addressing **SDG 9: Innovation**.	**BT 5**
CO6	Design and develop a detailed project report, advancing **SDG 17: Partnerships for the Goals** through collaborative and applied research.	**BT 6**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Field Data Collection and Database Development Execution of field survey and collection of primary and secondary data with proper guidance and recommendation from the Supervisor and the Department: • Household and Community Surveys • GPS Survey and Ground Truthing • Environmental and Socio-economic Data Collection • Data Coding, Classification and Database Preparation • Compilation of Secondary Data	12
Unit 2	Data Processing, Mapping and Analysis • Data Tabulation and Statistical Analysis • GIS Database Creation • Preparation of Thematic Maps • Graphs, Charts and Diagrams • Spatial and Temporal Analysis • Interpretation of Survey Findings Progress presentation of fieldwork and preliminary analysis.	12

Unit 3	Project Report Writing Preparation of dissertation/report in prescribed format: • Introduction • Review of Literature • Study Area Profile • Methodology • Results and Analysis • Discussion • Conclusions and Recommendations • References and Appendices	12
Unit 4	Final Presentation and Viva-Voce • Submission of Final Project Report • Preparation of PowerPoint Presentation • Research Communication and Presentation Skills • Final Project Presentation • Viva-Voce Examination	12
	Total	**48**

Note: Submission of project report in prescribed format and on specified date is mandatory. Equal weightages of marks for each stage of the work (upto 3rd stage) for internal evaluation of the project by the supervisor (60% of end semester examination). 40% of the total marks of end semester examination is for viva-voce and final presentation to be evaluated by an external examiner.

Text Books:

As per the list of given in syllabus based on topic selected

Reference Books:

As per the list of given in syllabus based on topic selected